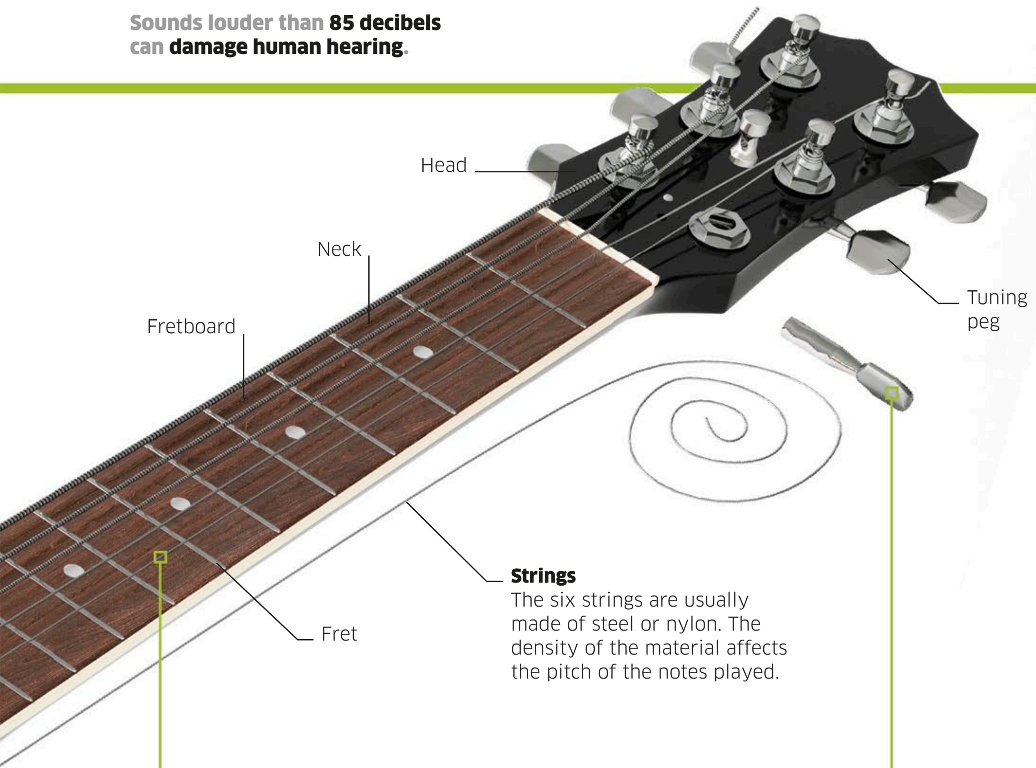
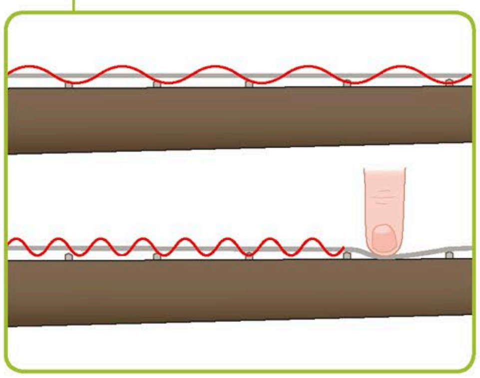
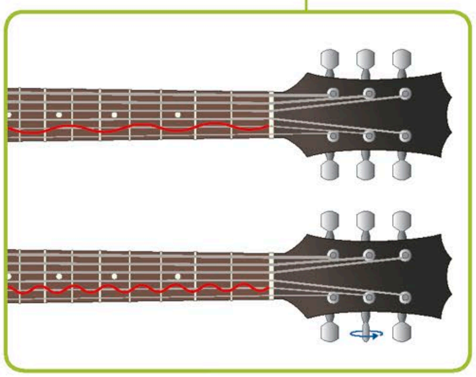


Strings
The six strings are usually made of steel or nylon. The density of the material affects the pitch of the notes played.



String length
Frets (raised bars) are spaced along the fretboard on the front of the neck. The player presses a string down on the fretboard to shorten its length, increasing the frequency and raising the pitch of the sound.



String tension
Turning the tuning pegs enables the player to tighten or loosen the strings, adjusting the pitch so that the guitar is in tune. As the strings are tightened, the frequency increases, raising the pitch.

Sound

Sound carries music, words, and other noises at high speed. It travels in waves, created by the vibration of particles within a solid, liquid, or gas.

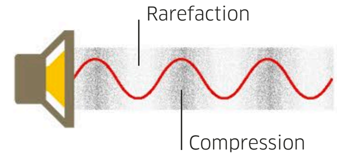
If you pluck a guitar string, it vibrates. This disturbs the air around it, creating a wave of high and low pressure that spreads out. When the wave hits our ears, the vibrations are passed on to tiny hairs in the inner ear, which send information to the brain, where it is interpreted. What distinguishes sounds such as human voices from one another is complex wave shapes that create distinctive quality and tone.

20 Hz to 20 kHz—the normal range of human hearing. This range decreases as people get older. Children can usually hear higher frequencies than adults.

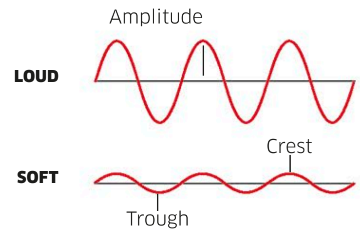
How sound travels

Sounds waves squeeze and stretch the air as they travel. They are called longitudinal waves because the particles of the medium they are traveling through vibrate in the direction of the wave.

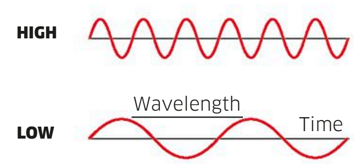
Vibrating particles
As vibrations travel through air, particles jostle each other to create high-pressure areas of compression and low-pressure areas of rarefaction.



Amplitude and loudness
The energy of a sound wave is described by its amplitude (height from center to crest or trough), corresponding to loudness.

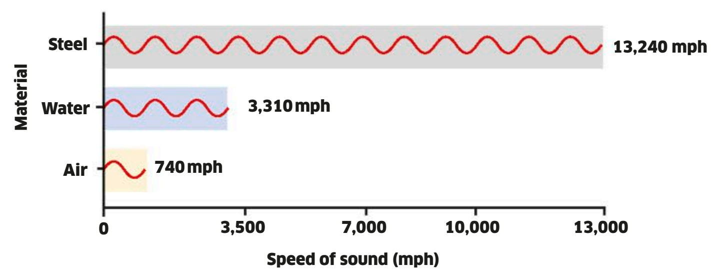


Frequency and pitch
A sound wave's pitch is defined by its frequency—the number of waves that pass a point in a given time. It is measured in hertz (Hz).



Speed of sound in different materials

Sound moves fastest in solids, because the particles are closer together, and slowest in gases, such as air, because the particles are further apart. The speed of sound is measured in miles per hour.



The decibel range

Loudness describes the intensity of sound energy, and is measured in decibels (dB) on a logarithmic scale, so 20 dB is 10 times more intense than 10 dB, or twice as loud. Human hearing ranges from 0 to 150 dB.

LEAF FALLING NEARBY (10 dB)
Barely audible

WHISPERING IN EAR (30 dB)
Quiet

SPEAKING NEAR YOU (60 dB)
Moderate

VIOLIN AT ARM'S LENGTH (90 dB)
Loud

FRONT OF ROCK GIG (120 dB)
Very loud

FIREWORK AT CLOSE RANGE (150 dB)
Painfully loud